

Khanyisa Mabasa

Midrand, South Africa

063 140 7110 | khanyimabasaa@gmail.com

Professional Profile

Motivated junior AI Software Developer with an NQF Level 5 qualification and hands-on experience building AI-powered web applications, interactive 3D prototypes, and browser-based game experiences. Skilled in Python, SQL, JavaScript, Git, Bubble.io, GDevelop, and frontend deployment tools. Recently completed a campaign game prototype combining GDevelop gameplay logic, Bubble.io website structure, leaderboard/database design, and HTML5 game embedding. Passionate about using technology, AI, and interactive experiences to solve practical problems and create engaging digital products.

Technical Skills

Programming & Development: Python, JavaScript, SQL, Object-Oriented Programming, Git, GitHub, Software Testing, Debugging, Clean Code, Documentation

AI & Machine Learning: Convolutional Neural Networks (CNN), Supervised Learning, Regression Models, Model Training and Evaluation, Feature Engineering, Data Preprocessing

Web & Interactive Development: HTML5, CSS basics, Three.js, API Integration, Frontend/Backend Integration, Responsive Web Prototyping

Game & No-Code Development: GDevelop, JavaScript Events in GDevelop, Game Logic, Timers, Scoring Systems, Collision Logic, HTML5 Game Export, Bubble.io, Bubble Workflows, Bubble Database Design, Leaderboards

Tools & Platforms: VS Code, Jupyter Notebook, Google Colab, Kaggle, Netlify, Render, GDevelop, Bubble.io

Experience

AI Software Developer

Retro Rabbit / Synergy Transformation Solution

June 2025 - May 2026

- Participated in structured AI and software development training aligned with QCTO standards.
- Built and tested machine learning models using Python.
- Applied software development best practices including Git version control.
- Practiced clean code principles, debugging, and technical documentation.
- Developed AI-powered prototypes combining user input, web interfaces, and interactive outputs.

Technical Support (Training Experience)

Makwa IT Technologies

September 2023 - August 2024

- Assisted users with hardware and software troubleshooting.
- Diagnosed system errors and supported issue resolution.
- Developed practical problem-solving, communication, and technical support skills.

Projects

CatchyMash - Checkers Sixty60 Campaign Game

- Built a mobile-friendly 60-second grocery-catching game for a Checkers Sixty60 campaign concept.
- Developed gameplay logic in GDevelop using JavaScript events, including falling items, random spawn positions, scoring, timer control, and same-scene game-over behavior.
- Implemented scoring rules for healthy items, treats, and negative items.
- Created an HTML5 playable game link using GDevelop publishing.
- Designed a Bubble.io prototype website structure with landing page, sign-up/login flow, embedded game page, manual score submission, and leaderboard.
- Created a Bubble database structure for users and scores, including leaderboard display logic.
- Prepared supporting campaign rationale and final submission documentation.

Playable link: <https://gd.games/games/b7ec46e4-b960-4047-b95f-f6fc632fdf18>

Bubble io:

https://catchymash-web-31017.bubbleapps.io/version-test/?debug_mode=true

Breast Cancer Detection Using CNN

- Built a Convolutional Neural Network model for breast cancer classification.
- Preprocessed medical image data for model training.
- Implemented training, validation, and performance evaluation.
- Improved classification performance through model optimisation.
- Demonstrated understanding of deep learning workflows and model evaluation.

AI-Powered 3D & Avatar Interaction System

- Developed AI-driven prototypes that convert text or image input into interactive 3D model experiences.
- Built an avatar interaction system that translates natural language commands into animated avatar actions.
- Integrated AI model outputs with a web-based frontend for interactive user experiences.
- Used Three.js for browser-based 3D rendering and dynamic visualization.

Project link: <https://ai-engineeringg.netlify.app>

Education

AI Software Developer - NQF Level 5

MICTSETA / QCTO

2026

Technical Support - NQF Level 4

MICTSETA

2024

Mechanical Engineering - N6

Sedibeng TVET College

2022

Reference

Mr Raymond

Synergy Transformation Solutions - Facilitator

084 456 9256